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BSIS02

1. Data Structures and Algorithms

Data Structures are ways of organizing and storing data in a computer so it can be used easily. Algorithms are step-by-step instructions used to solve a problem or complete a task.

Reference:
https://www.w3schools.com/dsa/dsa_intro.php

2. Linear Data Structure

A linear data structure stores data in a sequence, where the elements are arranged one after another. Examples include arrays, linked lists, stacks, and queues.

Reference:
https://www.w3schools.com/dsa/

3. One-Dimension Array

A one-dimensional array stores multiple values in a single list. Each value has an index. The first value starts at index 0. Arrays are useful for storing lists of related information.

Reference:
https://www.w3schools.com/dsa/dsa_data_arrays.php

4. Big O Notation

Big O Notation describes how much time an algorithm may need as the amount of data gets bigger. O(1) – stays the same, O(n) – increases with the amount of data, O(log n) – increases slowly. Big O helps us compare how efficient algorithms are.

Reference:
https://www.w3schools.com/dsa/dsa_timecomplexity_theory.php

5. Linear Search

Linear Search checks each item one by one until it finds the value being searched for. If we search for 30, it checks 10, then 20, then 30. Linear Search can be used even if the data is not sorted. Time Complexity: O(n)

Reference:
https://www.w3schools.com/dsa/dsa_algo_linearsearch.php

6. Binary Search

Binary Search searches for a value by checking the middle of a sorted list and removing half of the list each time. If we search for 40, it starts by checking the middle and then searches the correct half. Binary Search requires the data to be sorted. Time Complexity: O(log n)

Reference:
https://www.w3schools.com/dsa/dsa_algo_binarysearch.php